

Boss Challenge — Paper 50 (Class 7)

Weather, Climate & Adaptations | Nutrition in Animals | Integers | The Triangle & its Properties
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The day-to-day state of the atmosphere at a place - its temperature, humidity and rainfall right now - is called the:
(A) weather
(B) climate
(C) season
(D) humidity
2. Taking food into the body through the mouth is the step of nutrition called:
(A) ingestion
(B) egestion
(C) absorption
(D) digestion
3. The collection of whole numbers together with their negatives - ..., -3, -2, -1, 0, 1, 2, 3, ... - is called the:
(A) fractions
(B) integers
(C) decimals
(D) natural numbers
4. Add up the three interior angles of any triangle and you will always get:
(A) 270°
(B) 90°
(C) 360°
(D) 180°
5. Taken over about 25 years, the average long-term pattern of weather at a place is called its:
(A) temperature
(B) climate
(C) weather
(D) forecast
6. The breaking down of complex food into simpler, soluble substances the body can use is called:
(A) digestion
(B) egestion
(C) ingestion
(D) assimilation

7. Adding on the number line, $(-7) + 4$ comes to:
- (A) 11
 - (B) -11
 - (C) 3
 - (D) -3
8. Two angles of a triangle are 50° and 60° ; its remaining third angle measures:
- (A) 110°
 - (B) 70°
 - (C) 80°
 - (D) 90°
9. The instrument used to measure the temperature of the air is a:
- (A) anemometer
 - (B) rain gauge
 - (C) barometer
 - (D) thermometer
10. The wall of the food pipe and stomach pushes food forward by waves of muscle movement called:
- (A) digestion
 - (B) peristalsis
 - (C) absorption
 - (D) respiration
11. The value of $(-6) + (-5)$ is:
- (A) -1
 - (B) 1
 - (C) 11
 - (D) -11
12. A triangle with all three sides equal in length is called:
- (A) a scalene triangle
 - (B) an equilateral triangle
 - (C) a right triangle
 - (D) an isosceles triangle
13. The amount of water vapour (moisture) present in the air is known as:
- (A) rainfall
 - (B) temperature
 - (C) humidity
 - (D) pressure

14. The sharp, chisel-shaped front teeth used for biting and cutting food are the:
- (A) molars
 - (B) incisors
 - (C) canines
 - (D) premolars
15. The value of $8 - (-3)$ is:
- (A) -11
 - (B) 11
 - (C) 5
 - (D) -5
16. In an equilateral triangle, every one of the three equal interior angles measures:
- (A) 45°
 - (B) 60°
 - (C) 180°
 - (D) 90°
17. A polar bear survives the freezing Arctic mainly because of its:
- (A) thick fur and a layer of fat under the skin
 - (B) large ears that give off heat
 - (C) ability to store water in a hump
 - (D) thin skin with little fur
18. The pointed teeth next to the incisors, used mainly for tearing and piercing food, are the:
- (A) incisors
 - (B) molars
 - (C) premolars
 - (D) canines
19. The value of $(-4) - 6$ is:
- (A) 2
 - (B) -10
 - (C) 10
 - (D) -2
20. A triangle that has exactly two sides equal (and therefore two equal angles) is called:
- (A) a right triangle
 - (B) an isosceles triangle
 - (C) a scalene triangle
 - (D) an equilateral triangle
21. At dawn a hill station read -2°C ; by noon it had risen to 6°C . The rise in temperature was:
- (A) 6°C
 - (B) 4°C
 - (C) -8°C
 - (D) 8°C

22. A young child has 20 milk teeth. An adult has 32 permanent teeth. The number of extra teeth an adult has is:
- (A) 12
 - (B) 52
 - (C) 20
 - (D) 10
23. Multiplying a negative by a positive, $(-5) \times 3$ equals:
- (A) -15
 - (B) 15
 - (C) -2
 - (D) -8
24. By the exterior-angle property, an exterior angle of a triangle equals the sum of which angles?
- (A) opposite interior angles
 - (B) all three interior angles
 - (C) adjacent interior angles
 - (D) equal halves of the right angle
25. Birds like the Siberian crane fly thousands of kilometres to escape harsh winters. This seasonal journey is called:
- (A) camouflage
 - (B) migration
 - (C) hibernation
 - (D) adaptation
26. The watery juice in the mouth, called saliva, begins to digest:
- (A) water
 - (B) protein
 - (C) starch (carbohydrate)
 - (D) fat (oil)
27. When two negatives are multiplied together, $(-6) \times (-4)$ equals:
- (A) -24
 - (B) 24
 - (C) -10
 - (D) 10
28. In a triangle, an exterior angle measures 120° and one opposite interior angle is 70° . The other opposite interior angle is:
- (A) 120°
 - (B) 190°
 - (C) 50°
 - (D) 60°

29. The instrument that measures how much rain has fallen at a place is the:
- (A) hygrometer
 - (B) wind vane
 - (C) thermometer
 - (D) rain gauge
30. Which organ - the body's largest gland - produces the bile that helps digest fats?
- (A) small intestine
 - (B) pancreas
 - (C) liver
 - (D) stomach
31. Dividing a negative by a positive, $(-20) / 5$ gives:
- (A) 4
 - (B) -15
 - (C) -100
 - (D) -4
32. A line drawn from a vertex of a triangle to the MIDPOINT of the opposite side is called a:
- (A) altitude
 - (B) hypotenuse
 - (C) median
 - (D) bisector of a side
33. Animals of the hot, sandy desert, such as the camel, are mainly adapted to cope with:
- (A) deep shade and darkness
 - (B) very little water and great heat
 - (C) freezing cold and ice
 - (D) constant heavy rain
34. Most of the digested food is absorbed into the blood through tiny finger-like projections in the small intestine called:
- (A) alveoli
 - (B) villi
 - (C) nephrons
 - (D) cilia
35. The value of $(-18) / (-3)$ is:
- (A) -6
 - (B) 54
 - (C) 6
 - (D) -21

36. A perpendicular line drawn from a vertex of a triangle to the opposite side (the height) is called an:

- (A) exterior angle
- (B) hypotenuse
- (C) median
- (D) altitude

37. A tropical rainforest, found near the equator, has a climate that is:

- (A) mild with four equal seasons
- (B) hot and very dry
- (C) cold and snowy
- (D) hot and very wet (humid) all year

38. Animals such as cows and buffaloes that quickly swallow grass, then later bring it back to chew slowly, are called:

- (A) carnivores
- (B) decomposers
- (C) parasites
- (D) ruminants

39. The number that must be added to -9 to give 0 (its additive inverse) is:

- (A) 0
- (B) +9
- (C) 18
- (D) -9

40. In a right-angled triangle, the side opposite the right angle (the longest side) is called the:

- (A) hypotenuse
- (B) median
- (C) altitude
- (D) base

41. On three winter mornings a town recorded -4°C , 0°C and 4°C . The average (mean) of these temperatures is:

- (A) 0°C
- (B) -4°C
- (C) 8°C
- (D) 4°C

42. The part of the digestive system where digested food is finally used by body cells for energy and growth is the step called:

- (A) ingestion
- (B) egestion
- (C) absorption
- (D) assimilation

43. On the number line, of the two integers -3 and -7, the GREATER one is:
- (A) -7
 - (B) Neither can be compared
 - (C) -3
 - (D) They are equal
44. By the Pythagoras property, a right triangle with legs 3 cm and 4 cm has a hypotenuse of:
- (A) 5 cm
 - (B) 7 cm
 - (C) 12 cm
 - (D) 25 cm
45. The huge flat ears of an elephant help it mainly to:
- (A) store water for the dry season
 - (B) lose extra body heat and stay cool
 - (C) hear sounds from far underwater
 - (D) frighten away all predators
46. The removal of undigested, unabsorbed food as faeces from the body is the step called:
- (A) absorption
 - (B) egestion
 - (C) digestion
 - (D) excretion
47. The greatest negative integer is:
- (A) -1
 - (B) 0
 - (C) -100
 - (D) There is none
48. For any triangle, adding the lengths of any two sides always gives a total that is:
- (A) greater than the third side
 - (B) equal to the third side
 - (C) less than the third side
 - (D) equal to 180°
49. Which set lists only ELEMENTS of weather that a weather report describes?
- (A) rocks, soil and minerals
 - (B) temperature, humidity, rainfall and wind
 - (C) rivers, lakes and oceans
 - (D) plants, animals and forests
50. The single-celled animal Amoeba captures its food by pushing out finger-like extensions called:
- (A) cilia
 - (B) tentacles
 - (C) villi
 - (D) pseudopodia (false feet)

51. The integer that is neither positive nor negative is:

- (A) -1
- (B) 1
- (C) 0
- (D) There is no such integer

52. Which of these three lengths CANNOT form a triangle?

- (A) 5 cm, 6 cm, 9 cm
- (B) 7 cm, 7 cm, 7 cm
- (C) 3 cm, 4 cm, 8 cm
- (D) 6 cm, 8 cm, 10 cm

53. The polar regions are far colder than the equator chiefly because there the Sun's rays:

- (A) are blocked completely by clouds
- (B) never reach the polar ground at all
- (C) carry no heat energy near the poles
- (D) strike the ground very slantingly and spread out

54. Inside the stomach, the food is mixed with digestive juices that include a strong acid. This acid mainly:

- (A) cools the food down
- (B) makes the food sweet
- (C) adds water to the food
- (D) kills germs and helps the juices act

55. The predecessor of the integer -5 (the integer just before it) is:

- (A) -4
- (B) 6
- (C) -6
- (D) 4

56. A triangle in which one angle is exactly 90° is called:

- (A) an acute-angled triangle
- (B) a right-angled triangle
- (C) an obtuse-angled triangle
- (D) an equilateral triangle

57. A penguin colony huddles tightly together in the Antarctic mainly to:

- (A) hide from the bright sunlight
- (B) lay their eggs more quickly
- (C) share warmth and reduce heat loss
- (D) catch more fish on the ice

58. A full set of permanent teeth in an adult human numbers 32. If 8 of these are incisors, the number that are NOT incisors is:

- (A) 8
- (B) 40
- (C) 16
- (D) 24

59. The product of any integer and (-1) is:

- (A) the additive inverse of that integer
- (B) always 0
- (C) always -1
- (D) the same integer unchanged

60. In a right-angled triangle, one of the two non-right angles is 35° . The other non-right angle is:

- (A) 145°
- (B) 90°
- (C) 55°
- (D) 65°

61. At midnight a polar station read -9°C . By the next midday the temperature had risen by 5 degrees. The midday reading was:

- (A) -4°C
- (B) 4°C
- (C) -14°C
- (D) -5°C

62. Grass-eating animals can digest the tough cellulose of grass because their gut holds:

- (A) strong acid alone
- (B) helpful bacteria that break down cellulose
- (C) extra sets of teeth
- (D) no special help at all

63. Because $(-4) + (-7)$ gives the same result as $(-7) + (-4)$, addition of integers is said to be:

- (A) distributive
- (B) closure
- (C) commutative
- (D) associative

64. A triangle in which no two of the three sides are equal in length is called:

- (A) an isosceles triangle
- (B) an equilateral triangle
- (C) a right triangle
- (D) a scalene triangle

65. A red-eyed tree frog and a toucan with a long light beak are animals you would expect to find in a:
- (A) tropical rainforest
 - (B) grassland savanna
 - (C) sandy desert
 - (D) polar ice cap
66. After Amoeba engulfs its food, the food is digested inside a small bubble-like structure called the:
- (A) villus
 - (B) food vacuole
 - (C) nucleus
 - (D) gall bladder
67. The statement $3 \times (4 + (-2)) = 3 \times 4 + 3 \times (-2)$ shows the property called:
- (A) commutative
 - (B) distributive
 - (C) closure
 - (D) associative
68. In an isosceles triangle the two equal sides meet at the apex, and one base angle is 50° . The other base angle is:
- (A) 100°
 - (B) 80°
 - (C) 50°
 - (D) 130°
69. Over one week a city's noon temperatures had a highest reading of 9°C and a lowest of -3°C . The range (highest - lowest) was:
- (A) 9°C
 - (B) -12°C
 - (C) 12°C
 - (D) 6°C
70. The broad, flat back teeth used mainly for grinding and chewing food are the:
- (A) canines
 - (B) molars
 - (C) incisors
 - (D) milk teeth
71. A diver is at -15 m (15 m below sea level) and rises 9 m. Her new position is:
- (A) -24 m
 - (B) 6 m
 - (C) -6 m
 - (D) 24 m

72. A triangle that has one angle greater than 90° is called:

- (A) a right-angled triangle
- (B) an acute-angled triangle
- (C) an obtuse-angled triangle
- (D) an equilateral triangle

73. Animals living in tropical rainforests often have very specific features, such as a monkey's long gripping tail, because:

- (A) there is almost no life to compete with
- (B) there is no rainfall at all
- (C) the climate is freezing cold
- (D) competition for food and space is very high there

74. Bile, which helps digest fats, is made in the liver and stored for release in the:

- (A) pancreas
- (B) large intestine
- (C) gall bladder
- (D) stomach

75. The successor of the integer -1 (the integer just after it) is:

- (A) -1
- (B) 1
- (C) 0
- (D) -2

76. By the Pythagoras property, a right triangle with hypotenuse 13 cm and one leg 12 cm has the other leg equal to:

- (A) 5 cm
- (B) 1 cm
- (C) 17 cm
- (D) 25 cm

77. Which statement correctly contrasts weather and climate?

- (A) Climate refers only to rainfall, weather only to wind
- (B) Weather is the 25-year average, while climate changes each hour
- (C) Weather can change within a day, while climate is the long-term average over years
- (D) Weather and climate mean exactly the same thing

78. Tooth decay (cavities) is caused when bacteria act on left-over food and produce:

- (A) acids that damage the tooth
- (B) pure water that cleans the tooth
- (C) bases that strengthen the tooth
- (D) extra enamel on the tooth

79. Because the sum of any two integers is always again an integer, integers are said to be _____ under addition.
- (A) commutative
 - (B) open
 - (C) distributive
 - (D) closed
80. Every triangle has exactly three medians, and they all pass through a single common point called the:
- (A) right angle
 - (B) centroid
 - (C) vertex
 - (D) hypotenuse
81. A maximum-minimum thermometer is specially useful because it records the:
- (A) highest and lowest temperatures reached in a period
 - (B) amount of rain that fell overnight
 - (C) humidity of the air at sunrise
 - (D) speed of the wind during the day
82. The main job of the large intestine in digestion is to:
- (A) add acid to kill germs
 - (B) digest proteins fully
 - (C) absorb water from the undigested food
 - (D) make bile for fats
83. The value of $0 - 7$ is:
- (A) 7
 - (B) -14
 - (C) 0
 - (D) -7
84. An equilateral triangle, having all angles equal, is a special case of which other type?
- (A) isosceles triangle
 - (B) right triangle
 - (C) obtuse triangle
 - (D) scalene triangle
85. A desert fox has very large ears compared with an Arctic fox's small ears. The large ears mainly help the desert fox to:
- (A) hear sounds from underground only
 - (B) store extra food for winter
 - (C) swim quickly across rivers
 - (D) give off body heat and stay cool

86. Which sequence correctly orders the steps of nutrition in animals?

- (A) egestion -> assimilation -> absorption -> digestion -> ingestion
- (B) ingestion -> digestion -> absorption -> assimilation -> egestion
- (C) absorption -> ingestion -> digestion -> assimilation -> egestion
- (D) digestion -> ingestion -> egestion -> absorption -> assimilation

87. Which of these integers is the SMALLEST?

- (A) -12
- (B) 0
- (C) 7
- (D) -5

88. A triangle's three angles are in the ratio 1 : 2 : 3; the biggest of them measures:

- (A) 60°
- (B) 30°
- (C) 90°
- (D) 120°

89. The lion-tailed macaque, with its silver mane, lives in the rainforests of the:

- (A) Himalayan snow line
- (B) coastal mangrove swamps of Bengal
- (C) Thar Desert of Rajasthan
- (D) Western Ghats of India

90. The tongue helps in eating by mixing food with saliva, in swallowing, and also in:

- (A) sensing taste
- (B) making bile for digestion
- (C) producing the stomach's acid
- (D) absorbing digested food

91. Multiplying three signed numbers, $(-2) \times 5 \times (-1)$ comes to:

- (A) -10
- (B) 7
- (C) -7
- (D) 10

92. By the Pythagoras property, a right triangle with legs 6 cm and 8 cm has a hypotenuse of:

- (A) 10 cm
- (B) 48 cm
- (C) 14 cm
- (D) 100 cm

93. Penguins, polar bears and seals all share thick fat layers under their skin because in their habitat the most important need is to:
- (A) keep their body warm in extreme cold
 - (B) float upright on dry sand
 - (C) stay cool under a hot sun
 - (D) store water for a long drought
94. A cow first swallows grass into a part of its stomach called the rumen and later chews the returned food, which is known as:
- (A) saliva
 - (B) enamel
 - (C) cud
 - (D) bile
95. Grouping the numbers differently - $(-2 + 3) + 5$ gives the same sum as $-2 + (3 + 5)$ - illustrates the property called:
- (A) associative
 - (B) closure
 - (C) commutative
 - (D) distributive
96. A triangle in which all three angles are less than 90° is called:
- (A) a right-angled triangle
 - (B) an obtuse-angled triangle
 - (C) an acute-angled triangle
 - (D) a scalene triangle
97. On the same winter day a hill town read -6°C and a coastal town read 11°C . The coastal town was warmer than the hill town by:
- (A) 17°C
 - (B) 11°C
 - (C) -17°C
 - (D) 5°C
98. Bile produced by the liver is stored, until it is needed for digesting fats, in a small sac called the:
- (A) rumen
 - (B) pancreas
 - (C) appendix
 - (D) gall bladder
99. At dawn the temperature was -4°C and by noon it had risen to 9°C . The total rise in temperature was:
- (A) -13°C
 - (B) 5°C
 - (C) 36°C
 - (D) 13°C

100. In any triangle, the longest side always lies directly opposite the triangle's:

- (A) midpoint of the base
- (B) smallest angle
- (C) largest angle
- (D) right angle